Research vision. One concept I seek to explore is a **universal version control for personal information**. While software versioning primarily functions as a backup for commit history, I envision a concept that encourages revisiting one's digital past for reflection and sharing. In today's digital world, personal information is fragmented across external providers. **My goal is to restore individuals' autonomy to curate, organize, and selectively share any information that matters to them.** This reifies Ivan Illich's idea of "convivial society" where members are empowered to act autonomously and use tools to enrich their environment with the fruits of their vision in domains such as healthcare and misinformation [4]. My research focuses on designing intelligent systems that **scaffold online foraging and sensemaking** by leveraging behavioral signals, **curating resources into personal repositories automatically**, and **using visual interfaces to align user intentions with algorithms**, enhancing system explainability. With expertise and insights in Human-Computer Interaction (HCI), Programming Languages (PL), and Systems, I aim to pursue a Ph.D. to work towards this vision.

Explainable and automated online resources foraging. In my first project at the University of Michigan, advised by Dr. Xinyu Wang, I designed interactive interfaces that guided users to specify their information needs through demonstration. After web-scraping programs are synthesized via program synthesis, I mapped synthesized programs back to on-screen content with in-situ visualizations of predicted elements to scrape next for users to validate alignments between their intentions and the synthesized programs (published at PLDI 2022 [3]). In the follow-up work, advised by Dr. Tianyi Zhang and Dr. Xinyu Wang, we improved the explainability of the programs and enhanced users' confidence. Specifically, we introduced a mixed-initiative approach that allowed users to step through multiple synthesized programs with visual feedback, map programs to natural language descriptions, and iteratively correct misalignments by overwriting parts of the program through additional demonstrations (published at UIST 2023 [2]). These projects demonstrated that, with the right medium, even non-domain experts can effectively *express their information needs* and *validate alignment* between algorithms and their intent. From these explorations, I developed core HCI research skills, including prototyping and conducting user studies.

Passive foraging and restoring mental models during online sensemaking. At Brown, under Dr. Jeff Huang's guidance, I built a local personal web archive to help users refind what they once read online and restore their original mental models, even when web content changed over time. The dynamic nature of the web often alters important "waypoints" [1] along initial navigation paths and makes it difficult for people to "replay" initial navigation steps in information refinding. My solution directly points users to on-page informational anchors through full-text search, bypassing the error-prone orienteering [5] process during refinding. Specifically, it passively captures browser screenshots, extracts text, and indexes them for lookups. I conducted a mixed-method study consisting of 26 one-hour sessions, which revealed significant improvements in users' confidence during refinding and context restoration. This work culminated in the public release of IRCHIVER[1] and a first-authored submission to CHIIR 2025. Many user-study participants use IRCHIVER to track task-relevant data changes to refine strategies, suggesting a need for tools to (1) track diffs for any data they find important and (2) passively "remember" since users often cannot anticipate future information needs.

In an ongoing collaboration between CMU and Brown, advised by Dr. Aniket Kittur, Dr. Sherry Wu, and Dr. Jeff Huang, I explored ways to enhance collaborative sensemaking by helping team members understand each other's asynchronous online processes during task handoffs. I am developing a system that passively organizes unstructured information encountered during browsing into a structured digest for sharing. By leveraging *implicit and explicit browsing signals* and LLM reasoning, our system captures prominent options, key sensemaking factors, and screenshots augmented with search trails. It also highlights "roads not taken", options viewed but not pursued, either due to ongoing higher-priority tasks or a lack of interest. From this project, I learned that online sensemaking clusters information into conceptual

[1] The website for IRCHIVER.

tasks, driven by dependencies and conditions. This calls for systems to *infer conceptual tasks from behavioral signals and treat them as first-class entities in search and sensemaking.* From these experiences, I built on my foundational HCI skills and grew into a more independent researcher by leading efforts to identify gaps, formulate research questions, secure IRB approvals, design study probes, and analyze data.

Robust distributed systems. Additionally, I worked with Dr. Nikos Vasilakis at Brown in providing automated fault-tolerant scale-outs of shell scripts without any modification. We achieved this by transforming shell scripts into dataflow graphs through command annotations, lightweight communication primitives to track progress and ensure exactly-once semantics in fault recovery, and a dynamic orchestration system for efficient task (re-)scheduling based on dataflow dependencies. This effort led to a co-first authored paper submitted to OSDI 2025. In distributed systems, dependencies within dataflow graphs (lineage) are commonly used to efficiently recompute lost data and ensure fault tolerance [6]. The concept of "lineage" holds potential for *tracking provenance of data and mental models across application boundaries.* I also developed transferable skills in collaborating with co-leaders and designing, implementing, and evaluating data-intensive systems.

Why Ph.D.? My decision to pursue a Ph.D. stems from my personal and professional experiences. From 2015 to 2019, I volunteered as a translator on *Hupu.com*, China's largest sports forum, bridging language barriers by translating English basketball analytics articles into Mandarin. My proudest contribution was subtitling a defense masterclass by Jim Boeheim for a girls' primary school basketball team in Liangting'aoxiang, a resource-poor region in Hunan Province; the clip helped them win provincial tournaments. When *Hupu.com* removed all historic posts in 2019, I realized the fragility of centralized data control and the need for systems that enable individuals to preserve and organize meaningful information. At *Yerbba*, a medical informatics startup, I developed feature-based explanations for decision tree algorithms to help patients understand post-diagnosis options and assist developers in validating treatment logic. This work revealed that inconsistencies in Electronic Health Records—caused by conflicting doctors' phrasing—hinder patients from understanding their treatment options. These experiences inspired me to pursue a Ph.D. to gain the expertise needed to design and prototype systems that help users forage, organize, and make sense of their personal data. After completing my Ph.D., my goal is to pursue an academic career or launch a startup to build and deploy such "convivial tools" that restore autonomy in personal information.

Why UCSD CogSci? I look forward to working with three research groups at UCSD CogSci whose work aligns closely with my vision of empowering individuals and teams through innovative interfaces and systems while advancing concepts like universal version control for personal information. First, I am interested in Dr. Haijun Xia's work on augmenting human creativity with intelligent systems, especially transforming linear text into representations that fit human mental models. This inspires me to extend his approach to exploratory search, design, and coding, and eventually design "fluid" interfaces that allow systems to infer domain-specific mental models on a per-user basis, personalizing outputs to conform to individual representations. Second, I am intrigued by Dr. Steven Dow's work on building systems and designing interfaces for collective innovation. His research on understanding the creative process to improve design practices for teams aligns with my goal of enhancing team collaboration. For instance, I aim to explore a universal version control system for each team member's data, creating provenance of individual contributions and "intangible" mental models, which can be merged into a collective team insight—a "social beehive." Furthermore, Dr. Philip Guo's research on visualization-based scaffolding for learning programming interests me. I aim to build upon his work by augmenting version control systems to include annotations of each commit's mental model, adjusted for different expertise levels. By generalizing this approach beyond programming, I hope to facilitate learning at scale with universal version control of learners' mental models. These alignments make UCSD the ideal environment for me to advance my research on designing systems that enhance personal and collaborative information management through innovative, user-centric approaches.

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